**Harsh Patnaik**
B.Tech ECE | Systems Engineering, Hardware-Software Integration, HIL/SIL Validation
Bengaluru, India | harshpatnaik133@gmail.com | +91 8088271232 | LinkedIn | GitHub

## EDUCATION

PES University, Bengaluru | B.Tech in Electronics and Communication Engineering | 2024 – 2028 (Pre Final Year)

## TECHNICAL SKILLS

**Robotics & Autonomy Simulation:** ROS 2 (Humble), Gazebo (SITL), PX4, Betaflight, OpenCV, PID & Feedforward Tuning, Linux (Ubuntu CLI)
**CAD, Aerodynamics & Systems Analysis:** SolidWorks, Flow Simulation (CFD), Propulsion & Thermal Sizing, Kinematics
**Hardware, PCB & Lab Diagnostics:** Schematic & PCB Layout, Power Distribution, ESP32, Arduino, Sensors, Multimeter & Oscilloscope Diagnostics, Wiring & Soldering
**Communication & Fieldbus Protocols:** CAN Bus, ELRS RF, UART, I2C, SPI, WebSockets
**Languages & Engineering Tools:** Python (ROS 2 Nodes, Telemetry, Test Automation), C (State Machines & Pointers), HTML5 Canvas (Web Dashboards), EasyEDA, GNU Radio, PlotJuggler, ESC Configurator, Git, VS Code

## KEY PROJECTS & EXPERIENCE

**Advanced BVLOS Hybrid VTOL UAV (Patent Pending)** | Lead Systems Designer
*Technologies: Systems Engineering, PX4, Powertrain Architecture, Aerodynamics, SolidWorks, CFD, Kinematics, Conceptual Design*

- Engineered a 2m BWB in SolidWorks; validated a ~13.0–13.7 cruise L/D across a 24-point CFD study and core shear safety (FoS = 2.69).
- Designed a phase-specific thrust-vectoring propulsion, modelled a propulsion allocation matrix with blade-element stall derating, established kinematic torque balancing and strict stator thermal constraints, validated actuator mixing and control allocation across VTOL transitions in PX4 SITL with Gazebo.
- Specified onboard avionics, dual-phase cooling, and BVLOS telemetry with GNSS fallback logic and a ballistic parachute failsafe (5.4 m/s fall).
- Validated BVLOS mission margins (Sim-to-Real drag buffers, 10% weather penalties, 80% battery reserves) and conceptualized rib-segmented BWB packaging for single-person logistics.
- Minimized VTOL transition losses to achieve >2x higher payload (8 - 9 kg) and >1.5x range (150+ km) over comparable 2m-wingspan VTOLs

**e-Yantra Robotics Competition (eYRC 2025) – Stage 2 Qualifier** | Robotics & Drone Engineer
*Technologies: ROS 2 Humble, Python, Betaflight, Gazebo, ELRS Hardware*

- Integrated an indoor ROS 2 (Python) UAV testbed combining overhead-camera localization with an ELRS RF bridge to a Betaflight drone, tuning velocity damping and voltage feedforward
- Built a non-blocking ROS 2 Action Server/Client framework supporting concurrent navigation, real-time telemetry, and stabilization.
- Developed an overhead camera OpenCV pipeline utilizing adaptive colour thresholding for robust plant and asset detection under variable lighting.
- Analysed telemetry to tune filters and PWM ramping, mitigating oscillations and current spikes.
- Resolved ESC mismatches, designed compact 555-timer latching circuits with LED feedback for reliable pesticide payload state retention.

**Vega Racing Electric** | Electronics & Hardware Integration Engineer
*Technologies: Embedded Systems, Data Acquisition, LV Architecture, Wiring Harness, Telemetry, PCB Design*

- Collaborated on building and wiring the Low-Voltage (LV) distribution box; performed multimeter continuity mapping, pinout verification, isolation checks.
- Designed the schematic and PCB layout for a 24V STM32F407 steering controller with dual H-bridges, CAN transceivers, watchdog, and encoder pinouts.
- Collaborated on the vehicle telemetry data-acquisition architecture and real-time driver dashboard interface to aggregate subsystem sensor streams.

**BVLOS Autonomous Rover & Fault-Tolerant GCS Testbed |** Systems & Simulation Project
*Technologies: ROS 2 Humble, Gazebo, LiDAR Localization, Weighted Dijkstra, WebSockets, HTML5 Canvas*

- Built a split-compute Gazebo and ROS 2 rover simulation decoupled from a WebSocket HTML5 Ground Control Station.
- Configured a Python Weighted Dijkstra path-planning node with line-of-sight waypoint smoothing and terrain-scaled velocity limits across multi-cost zones
- Programmed a GPS-denied fallback node in Python that clusters 180-sample LiDAR scans against 12 fixed landmarks during simulated GNSS dropouts
- Engineered fault-tolerant telemetry that sheds payload data at 30% packet loss and buffers 250 path coordinates during cellular blackouts.

## OTHER PROJECTS

- **Recon Bot – Hackezee Hackathon 2025 (Runner-Up):** Prototyped a teleoperated hazard-zone robot in 8 hours featuring real-time sonar mapping, gas detection, and a cloud-based IoT sensor dashboard.
- **COMS-1 Satellite Weather Telemetry Decoder (GNU Radio, Python):** Constructed a baseband signal-processing flowgraph in GNU Radio to demodulate and decode recorded .wav satellite downlink captures into meteorological imagery which was further analysed for insights on weather.
- **Terminal-Based Adventure Runner Game**: Developed a procedural C game using state machines, dynamic memory allocation, function pointers, and binary file storage for persistent player progression.

## AREAS OF INTEREST

Systems Engineering, Autonomous Systems, UAV Dynamics & Control, Embedded Systems, EV Architecture, Robotics Simulation, Computer Vision